# Progress Report 4

1. **Project title:**
   Swarm of robots to complete locomotive and object-searching tasks, and touchscreen interface for a multipurpose robot (R1D4).

2. **Names, student numbers, and emails:**
   Group 36 members: (all members are registered in ECSE 458-D1 and ECSE 458-D2)


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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**
   All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

   Weekly meeting with the entire group: Wednesdays at 3:00 pm (previously it was Tuesdays at 3:45pm)
   Weekly meeting with the advisor: Thursdays at 4:30 pm

1. **Recent Progress:**

We finished the frontend of the application. The communication from the telnet server and to the application has been established. We are underway in establishing the connection the other way round. Recently we have almost finished development on the R1-D4 tablet project. The only thing that is missing is figuring out a way to break down the schedule file into small enough bytes that we can transmit it over the USB connection. Most likely we will convert it to some sort of string and then send it in 4 byte chunks. We have also started on the swarm robots project. Recently, a new Swarm Robot has arrived so we are figuring out how to add that one into the mix or if we should do it all and instead focus on improving current functionality.

2. **Future Plans:**

We are going to schedule a meeting with Prof. Vybihal to get all the equipment for the swarm robots so we can start working on it. As soon as we get that, we will assign clear goals and milestones for the project after familiarising ourselves with the working of the robots. Some of the goals outlined so far are to fix some components/functions and to be able to give the system the ability to send a command from the server to all the robots and group multiple such commands together. We would also divide up the tasks best suited to each team member's capabilities. Meanwhile, we also want to start preparing for the upcoming midterm presentation. We want to have a basic outline of the presentation slides completed within the next 2 weeks so that we are prepared in advance for it.

3. **Lifelong Learning:**

### a. What We Need to Learn for the Rest of the Semester

To successfully complete the R1D4 robot's touchscreen interface, we need to develop a better understanding of troubleshooting and problem-solving within an embedded system environment. Our current challenge is ensuring smooth communication between the tablet and the ESP32, which requires a systematic approach to identifying and resolving issues. This involves carefully testing different ways of transmitting data, making sure that the messages sent from the tablet are properly received and understood by the ESP32.

On top of technical challenges, we need to develop a better understanding of project management and collaboration. Properly dividing responsibilities, communicating well, and setting realistic milestones are crucial in making progress in a timely manner. The ability to balance individual workloads without compromising team objectives is one that will be used long after this project.

Another important area of growth would be interdisciplinary communication. Our project requires consistent and effective verbal and written communication among team members, our advisor, other capstone groups, and potential end users. Keeping clear documentation of our progress, challenges, and solutions ensures that everyone stays aligned throughout the project. Regular meetings, detailed progress reports, and structured discussions help in maintaining transparency and efficiency. Additionally, ensuring that messages are conveyed concisely and professionally—whether in emails, reports, or presentations—enhances collaboration and minimizes misunderstandings, ultimately contributing to the project's success.

Also, considering human factors and usability is essential in our project. The Android tablet interface should be intuitive and accessible to users with varying levels of technical expertise. By incorporating user feedback and refining the interface's layout, we can aim towards the goal of usability and efficiency. Ensuring large, clear buttons and responsive design, such as the robot's

different phase animation on the homepage, will make it easier for users to interact with the system without confusion.

Moreover, working on this project has really highlighted how important teamwork, problem-solving, and adaptability are. We've had to handle unexpected challenges, rethink our approach when things didn't work, and stay flexible in adapting to new information. Staying organized, communicating effectively, and being open to feedback have played a big role in keeping things moving forward. We've also learned how crucial it is to follow industry regulations and safety standards, especially since this robot is designed for a hospital environment.

### b. Takeaways on Self-Learning

One of the biggest crucial lessons that we have learned throughout the first part of our Capstone project is that experimentation is just as important as theoretical knowledge. Understanding concepts along with actually testing things in practice has given us a much deeper understanding of how everything in our project works. Debugging, in particular, has taught us that trial and error is an essential part of learning. By systematically testing different approaches, we've been able to identify what works, refine our methods, and make progress in the project.

Collaboration has also played a huge role in learning new skills. By working together and using each team member's strengths, we have been able to solve problems much faster than if we were working alone. Talking through issues with our advisor and other capstone groups has given us fresh perspectives and often led to solutions we wouldn't have considered on our own.

Finally, adaptability has been very significant. We initially planned to develop a website-based interface, but as we progressed, it became clear that an Android app would be a better fit for our project. Being flexible enough to reconsider our decisions and adjust our approach has been very important in making sure we build something that actually works well in practice. This experience has reinforced the importance of staying open to new ideas and adapting as needed, which is a valuable mindset for any project in the engineering field.